

Financial Value and Commercial Viability

Unit economics, five-year operating case, capital requirement and investor interpretation

Investor conclusion Blockmediary has credible high-margin workflow economics, but commercial distribution and regulatory execution - not the testnet transaction logic - determine whether that potential becomes a real business.

£3.06m	Month 12	£1.37m	Y4 M8
Initial operating funding	First paid pilot	Base Year-3 revenue	Monthly cash-flow break-even

Prepared by Blockmediary project team

Master's-level preliminary business report for academic and investor review

Reference model: Blockmediary Financial Model, 13 August 2026

Status: preliminary planning report; not an audited forecast, valuation or investment recommendation

Executive conclusion

Blockmediary's potential real value is financial as well as technical. The product replaces an all-or-nothing bilateral payment decision with prefunded, document-conditioned settlement. At the model's Year-1 deal sizes, the standard fee is £305 for a £35,000 Tier A eBL workflow, £540 for a £30,000 Tier B carrier/API workflow and £1,200 for a £30,000 Tier C paper workflow. Those fees equal 0.9%, 1.8% and 4.0% of deal value respectively, before any conditional dispute fee. The customer is paying for payment assurance and document verification; Blockmediary is not lending, taking FX risk or earning a spread on client money.

The base case reaches £76.0m of Year-3 gross merchandise value (GMV) across 2,000 deals. It converts that throughput into £1.371m of revenue, £1.057m of gross profit and a 77.1% gross margin, but still records negative EBITDA of £409k. The model crosses monthly cash-flow break-even in Year 4, Month 8, after a step from 2,000 to 15,000 annual deals and from 333 to 1,875 active customers. That is the central investment trade-off: strong unit economics are visible before company profitability, but the base case needs substantial distribution scale to absorb payroll, fixed operating cost and regulatory overhead.

Financial judgement The project is fundable as a milestone-gated pilot case, not yet as a forecast-backed scale case. The initial operating requirement is £3,063,210; the low case needs £4,279,063 and remains below monthly break-even beyond Month 60.

What this report adds

The BEEM063 brief places 70% of the group presentation mark on potential real value and asks the team to separate what has been built from the longer business plan. This report answers that test through the workbook: transaction economics, value capture, operating leverage, capital need, downside exposure, stage gates and the evidence required before the forecast can be trusted.

Companion document	It owns	This report uses it for
Legal and Compliance	Permissions, client-asset, AML and data risks	Cost, cash timing and investment gates
Deal Value Research	Evidence for £35k / £30k / £30k Year-1 averages	Revenue base and sensitivity implications
DIFC / VARA Readiness	Regulatory route and activity mapping	Month 12 / Month 18 timing and cost treatment
Competitor Analysis	Market structure and fee comparables	Pricing discipline and value-capture interpretation
Trade Escrow Agreement	Release, objection and dispute mechanics	Which events create fees and operating cost

Sources: BEEM063 Hackathon Main Presentation Brief 2026, p.1; companion submission PDFs listed in References.

1. What the model says about real-world value

The financial model treats trust as a workflow product rather than a credit product. The buyer prefunds the agreed stablecoin amount; the seller sees that funds are locked before shipping; release follows documented compliance and the objection process. This changes who bears timing and counterparty risk without requiring Blockmediary to advance capital against the trade.

Stakeholder	Economic problem	Financial mechanism	Value interpretation
Buyer	Avoid paying the seller outright before shipment evidence exists	Funds remain locked pending document conditions	The service fee is small relative to the trade principal, but it is not an insurance guarantee
Seller	Avoid shipping with only an unsecured promise of later payment	Prefunding is visible before shipment; valid release does not depend on discretionary buyer approval	Economic value is faster, more certain access to agreed funds when documents comply
Blockmediary	Monetise trust without lending or balance-sheet credit exposure	Escrow, document, exception and partner/API fees	Base total revenue yield falls from 2.7% to 1.2% of GMV as scale and digital mix increase
Partner / regulator	Operate a controlled and auditable settlement perimeter	Client assets are gross memorandum balances with matching liabilities	0% is available to operations and no client-fund yield is recognised

Sources: Financial Model, P&L 5yr!C7:G31 and Client Funds Memo!A5:H31; Trade Escrow Agreement, pp.2-4 and 18-19.

How much trade value is supported by each pound of revenue?

In the base case, the platform processes 55.4 pounds of Year-3 GMV for every pound of company revenue. The multiple expands to 80.6x by Year 5 because the digital tier gains share and the blended escrow take rate declines. This is throughput leverage, not balance-sheet leverage: GMV belongs to customers and is never company revenue or operating cash.

55.4x	1.8%	77.1%	£2.92m
Y3 GMV / revenue	Y3 total revenue yield	Y3 company gross margin	Y3 avg safeguarded assets

Boundary GMV, client escrow assets and future VARA restricted capital are not company cash. The model gives all three zero operating availability.

2. Economics of one transaction

Price formula Standard fee = max(deal value x tier take rate, minimum escrow fee) + document/review fee. Conditional dispute or amendment fees are excluded until the event occurs.

Tier	Workflow	Y1 deal	Take rate	Standard fee	Fee / deal	Conditional event fee
Tier A	eBL	£35,000	0.8%	£305	0.9%	£100
Tier B	carrier/API	£30,000	1.5%	£540	1.8%	£200
Tier C	paper	£30,000	3.0%	£1,200	4.0%	£500

Source: Financial Model, Assumptions!F63:F74 and F149:F154; Tier Economics!B3:U18. Standard fee excludes any dispute/amendment event.

Contribution economics before shared platform costs

Tier	Expected revenue / deal	Tier-specific variable cost	Contribution / deal	Margin before shared COGS
Tier A	£309	£10.4	£298.6	96.6%
Tier B	£560	£48.0	£512.0	91.4%
Tier C	£1,300	£390.0	£910.0	70.0%

Source: Financial Model, Tier Economics!B15:D26. Expected revenue includes probability-weighted dispute/amendment fees; costs exclude shared COGS, payroll and fixed overhead.

Tier A is the strategic scale engine. At the Year-1 assumptions, a £305 standard customer fee supports a £35,000 trade. The expected revenue of £309 and tier-specific variable cost of £10.40 produce a 96.6% margin before shared platform cost. This does not mean the company is 96.6% profitable: Year-1 company gross margin is 71.1% after shared COGS, and EBITDA remains negative after payroll and fixed operating expenditure.

Tier B is the balanced operating case. It carries more documents and human/API handling than Tier A while retaining a 91.4% margin before shared COGS. Its £540 standard fee equals 1.8% of the modeled £30,000 deal value.

Tier C is a priced fallback. The paper workflow has the highest fee, highest expected dispute rate and highest cost to serve. It remains economically positive, but its 70.0% margin before shared COGS is materially below the digital tiers. The scale plan therefore reduces its mix rather than assuming paper processing can remain the dominant workflow.

Validation requirement These prices are management hypotheses informed by the deal-value and competitor reports. They are not an observed tariff or proof of willingness to pay. The paid pilot must test conversion by tier, deal value and fee.

3. Digital mix creates operating leverage

Digital document mix expands while paper becomes a fallback

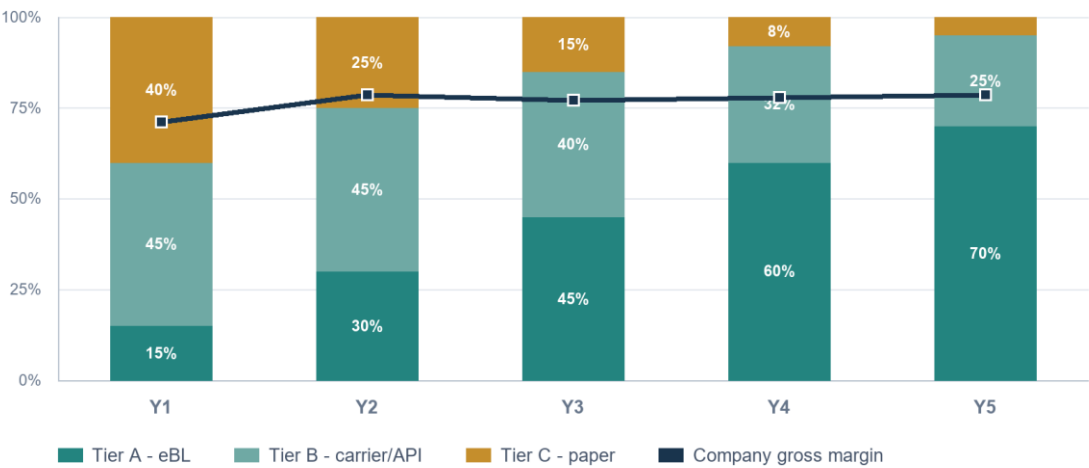


Figure 1. Base-case tier mix and company gross margin.

Source: Financial Model, Tier Economics!B3:U26 and P&L 5yr!C29:G31.

Metric	Year 1	Year 3	Year 5	Commercial meaning
Tier A share	15%	45%	70%	eBL becomes the dominant scale path
Tier C share	40%	15%	5%	paper remains available but becomes exceptional
Blended escrow take rate	2.0%	1.3%	1.0%	lower pricing accompanies digitisation
Documents per deal	6.75	5.10	4.05	weighted handling load declines
Margin before shared COGS	78.1%	86.9%	93.1%	mix and automation improve contribution economics
Company gross margin	71.1%	77.1%	78.6%	shared risk and operating costs keep the margin realistic

Source: Financial Model, Tier Economics!E4:U26 and P&L 5yr!C31:G31.

The model does not create margin by keeping prices high. It creates margin by shifting work into a lower-cost document pathway. That lets the blended escrow take rate fall from approximately 2.0% to 1.0% while whole-company gross margin stays within 71% to 79%. For customers, the strategic promise is cheaper, faster digital handling. For investors, the requirement is that the eBL/API adoption curve actually occurs.

4. Five-year operating case

Base-case revenue and EBITDA trajectory

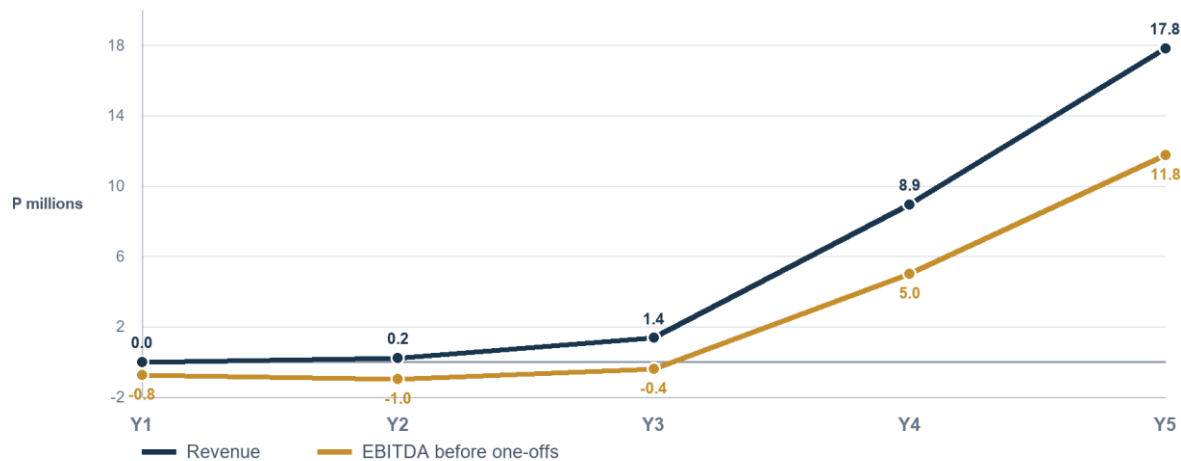


Figure 2. Base-case revenue and EBITDA trajectory.

Source: Financial Model, P&L 5yr!C18:G18 and C61:G61.

Year	Deals	Active customers	Avg deal	GMV	Revenue
Year 1	3	1	£30,750	£92.2k	£2.45k
Year 2	240	60	£34,005	£8.16m	£220k
Year 3	2,000	333	£38,016	£76m	£1.37m
Year 4	15,000	1,875	£42,823	£642m	£8.94m
Year 5	30,000	3,000	£47,897	£1.44bn	£17.8m

Source: Financial Model, P&L 5yr!C7:G18 and C69:G84. Active customers are model-implied targets, rounded for presentation.

Year	Gross profit	Gross margin	EBITDA before one-offs	Operating result	Cumulative result
Year 1	£1.74k	71.1%	(£764k)	(£1.11m)	(£1.11m)
Year 2	£173k	78.5%	(£990k)	(£990k)	(£2.1m)
Year 3	£1.06m	77.1%	(£409k)	(£409k)	(£2.51m)
Year 4	£6.96m	77.8%	£5m	£5m	£2.49m
Year 5	£14m	78.6%	£11.8m	£11.8m	£14.2m

Source: Financial Model, P&L 5yr!C29:G66. The model is pre-tax and pre-financing; corporation tax, interest and depreciation/amortisation are not modeled.

The Year-3 bridge: proof of monetisation, not yet profitability

Year-3 revenue stream	Amount	Share of total
Escrow transaction fees	£1,022,499	74.6%
Launch discounts and credits	(£69,871)	-5.1%
Document and exception-handling fees	£234,100	17.1%
Partner / API and ancillary services	£184,560	13.5%
Total revenue	£1,371,288	100.0%

Source: Financial Model, P&L 5yr!E14:E18.

Year 3 demonstrates a diversified revenue engine: escrow fees remain the majority, while document/exception and partner/API revenue contribute meaningful value. Yet £1.057m of gross profit does not fully cover £1.015m of payroll and £451k of fixed operating expenditure. The business therefore needs either more deals, more revenue per active customer or lower fixed-cost growth before it becomes self-funding.

The Year-4 inflection is the largest assumption in the model

Scale test From Year 3 to Year 4, deals rise 7.5x, active customers rise 5.6x and revenue rises 6.5x. Gross new customers required per Sales/BD/CS FTE rise from 93 to 225, against a model warning threshold of 250.

The model is internally consistent - the commercial capacity check still reads within capacity - but capacity is not demand. Partner distribution, self-serve onboarding, digital-document adoption and an AI-enabled operating model must all work together. The Year-4 and Year-5 outputs should therefore be read as targets conditional on those capabilities, not as a straight-line forecast from the prototype.

5. Capital requirement and use of funds

The base case requires £3,063,210 of initial operating capital. This equals the modeled peak monthly operating deficit of approximately £2,871,700 plus a three-month operating reserve of £191,516. Future VARA restricted capital of £319,095 is separate and cannot be used as runway.

Initial operating funding use

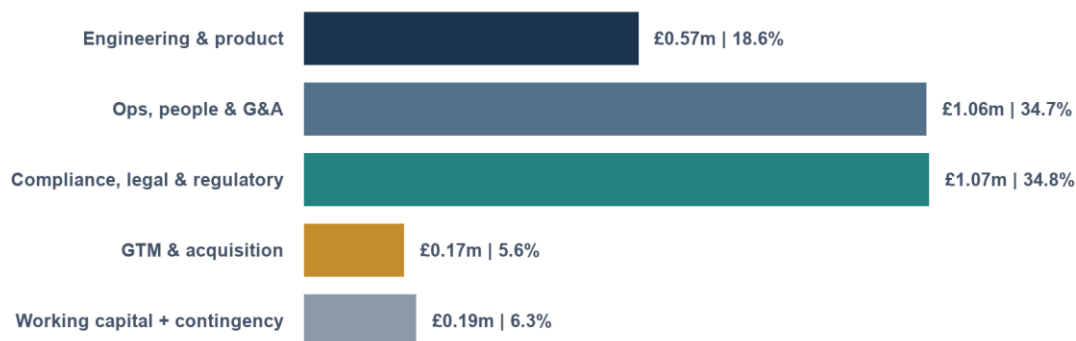


Figure 3. Initial operating funding allocation.

Source: Financial Model, Dashboard!J57:M62.

Capital item	Amount	Treatment
Peak operating cash gap	£2,871,700	Maximum cumulative deficit before financing
Three-month operating reserve	£191,516	Included in initial operating funding
Initial operating funding	£3,063,210	Spendable operating runway
Future VARA restricted capital	£319,095	Separate; not operating cash
Combined staged capital envelope	£3,382,305	Derived total; not the initial ask

Source: Financial Model, Launch Readiness!E12:E17; P&L 5yr!F64:F66; Dashboard!J57:M62.

What the capital is buying

People and permission dominate the ask. Operations, people and G&A account for 34.7%; compliance, legal and regulatory work account for 34.8%. Together they are 69.5% of the operating requirement. Engineering and product are 18.6%, go-to-market 5.6%, and reserve/contingency 6.3%. The working prototype reduces product-feasibility uncertainty, but production security, key management, partner integration and a full audit remain real engineering work.

Cost item	Model value	Financial treatment
First-six-month startup build	£243,412	Includes 15% contingency
DIFC / regulated-partner pilot onboarding	£60,000	Year-1 one-off

Cost item	Model value	Financial treatment
VARA application placeholder	£42,546	Year-1 staged application cost
Total Year-1 startup and staged regulatory one-offs	£345,958	Modeled operating expense
Partner-pilot oversight	£67,000	Annual pilot-stage run-rate
VARA supervision placeholder	£102,110	Annual cost after licensing

Sources: Financial Model, Startup 6mo!F40:F44 and Launch Readiness!A5:E17. VARA figures are planning placeholders pending counsel and regulator confirmation.

Reserve sensitivity

Reserve policy	Operating reserve	Funding need
3 months	£0.2m	£3.1m
6 months	£0.4m	£3.3m
9 months	£0.6m	£3.4m

Source: Financial Model, P&L 5yr!C89:E91. Rounded presentation values.

6. Downside, base and upside

Cash runway diverges sharply after Year 3

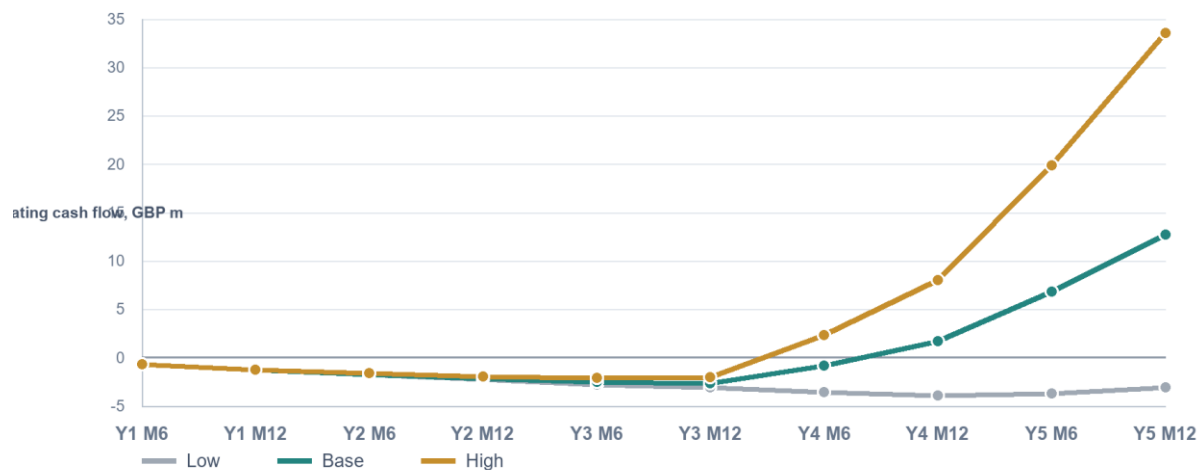


Figure 4. Six-month cumulative operating cash-flow checkpoints by scenario.

Source: Financial Model, Dashboard!B67:E77 and Cash Flow Statement!B45:F47.

Case	Y3 revenue	Y3 EBITDA	Y3 active customers	Monthly break-even
Low	£1.04m	(£748k)	480	Beyond Month 60
Base	£1.37m	(£409k)	333	Y4 M8
High	£1.91m	£100k	320	Y4 M4

Source: Financial Model, Scenario Engine!D30:P62 and Cash Flow Statement!B45:B47.

Case	Peak cash gap	Initial operating funding	Month-60 operating cash flow
Low	£4.088m	£4.279m	(£3.104m)
Base	£2.872m	£3.063m	£12.717m
High	£2.264m	£2.455m	£33.596m

Source: Financial Model, Cash Flow Statement!B45:F47 and Dashboard!J23:N28.

The low case is not a modest miss. It requires £4.279m of operating funding, does not reach monthly break-even inside five years and ends Month 60 with cumulative operating cash flow of negative £3.104m. This is the economically relevant downside because payroll and fixed operating costs do not shrink automatically when commercial volume misses.

The base case is a scale case. It reaches monthly break-even in Year 4, Month 8 and ends Month 60 at positive £12.717m cumulative operating cash flow before financing. That outcome is driven by the Year-4 volume step, not by aggressive pricing.

The high case proves operating leverage. It reaches positive Year-3 EBITDA and monthly break-even in Year 4, Month 4, with a lower £2.455m initial operating requirement. It should be read as upside capacity, not as the fundraising base.

Scenario caveat Low, Base and High vary deal counts and tier mix while payroll, fixed opex and one-offs remain fixed. They are planning cases, not probability-weighted outcomes.

7. Stage reached versus the longer plan

The delivered product is a working proof of concept on Base Sepolia testnet, not a live regulated service. It includes a non-custodial escrow contract, structured deal intake, deterministic document-rule grading, a release and refund path, an objection window, multi-deal dashboards and an audit trail. Production mainnet deployment, independent security audit, hardware-secured release keys, live KYC/KYB/sanctions screening, AI/OCR extraction and full regulatory permissions remain roadmap work.

Dimension	Reached now	Required before real-value scale
Settlement	Escrow create, fund, release and refund on Base Sepolia with test USDC	Mainnet/token approval, custody/control opinion, production key management
Document verification	Deterministic rules engine; structured bill-of-lading data	AI/OCR extraction, wider document set, human review console
Compliance	Requirements, risk register and legal template designed	Live KYB/KYC/sanctions controls, partner sign-off and licensing
Commercial	Pricing, deal values, tier economics and five-year model	Observed willingness to pay, paid pilot conversion, churn and deal frequency
Financial	42 control checks pass and scenarios reconcile	Actual revenue, cost-to-serve, holding period and acquisition performance

Sources: repository README and Business Requirements Document; DIFC / VARA Readiness Report, pp.2-3 and 7-11; Legal and Compliance Risk Assessment, p.2.

Commercial and regulatory timeline

Timing	Milestone	Financial meaning
Now	Working testnet prototype	Technical feasibility and deterministic release workflow demonstrated
Month 1	DIFC partner-pilot and VARA workstreams begin	Counsel, partner perimeter and application work start in parallel
Month 12	First paying partner-led pilot	First external evidence for price, conversion and operating cost
Month 13	First cash receipt	One-month collection lag modeled
Month 18	Full VARA target - base timing	Wider Dubai scale route, subject to permission scope and approval
Y4 M8	Monthly cash-flow break-even	Base case only; depends on the Year-4 distribution step

Sources: Financial Model, Launch Readiness!A19:J23 and Cash Flow Statement!B45:B47; DIFC / VARA Readiness Report, pp.1-3 and 11.

Stage statement for submission Blockmediary has demonstrated a working testnet escrow and deterministic document-release workflow. The financial model describes the funded route from that prototype to a paid partner pilot and licensed operation; it does not claim current revenue, current customers or permission to transact with real funds.

8. What must be proven before scale capital

The model is most useful as a measurement plan. Each unobserved driver can be converted into a pilot metric, and each metric can become a funding gate. This is more credible than treating five-year outputs as a valuation.

Driver to validate	Minimum evidence	Model input to replace
Deal value distribution	15-20 target SME interviews plus anonymised invoices; ideally at least 200 recent UAE forwarder jobs	Median, quartiles, right tail and share above £50k by tier
Willingness to pay	20-30 target-user price tests and paid-pilot conversion	Conversion probability by tier, deal value and fee
Document operations	Time-and-motion study with documentary reviewers	Minutes per pack, rework, discrepancy, escalation and loaded cost
Commercial motion	Pilot cohort telemetry	Acquisition source, CAC, churn, deals per active customer and revenue per customer
Settlement operations	Pilot transaction telemetry	Funded-to-release days, failure rate, objection rate and support burden
Regulatory perimeter	Integrated DFSA / VARA / CBUAE memorandum plus partner responsibility matrix	Permission scope, entity design, capital, fees and launch legality

Sources: Deal Value Research Report, pp.24-25; DIFC / VARA Readiness Report, p.11; Financial Model, Assumptions, Tier Economics and Client Funds Memo.

Recommended financing discipline

Funding gate	Evidence required	Why it protects capital
Gate 1 - perimeter and production readiness	Integrated UAE perimeter advice, named regulated partner, responsibility matrix, production security and audit plan	Confirms that the proposed pilot can legally and safely generate evidence
Gate 2 - paid pilot	Signed pilot customer, measured price acceptance, document workload, funded-to-release time and exception rate	Replaces the highest-impact unobserved unit-economics assumptions
Gate 3 - repeatability	Repeat customers, partner channel evidence, stable gross margin and conversion by tier	Tests the Year-3 customer and deal-frequency bridge
Gate 4 - scale and licensing	Application progress, distribution capacity, product automation and control evidence	Supports the Year-4 volume step and wider licence investment

Interpretive recommendation based on the financial model and companion research; the workbook does not prescribe funding tranches.

Decision rule Do not release scale capital merely because technical milestones are complete. The paid pilot must replace price, workload, conversion and cycle-time assumptions with observed data.

9. Model integrity, definitions and limitations

Mechanical status PASS: 42 of 42 model checks pass, and the formula-error scan found 0 cached #REF!, #DIV/0!, #VALUE!, #NAME?, #N/A or #NUM! errors.

A PASS means the workbook reconciles mechanically: the scenario selector, P&L roll-ups, cash roll-forward, revenue and COGS ties, client-fund memorandum, use-of-funds bridge, launch timing and restricted-capital treatment all agree. It does not mean that customers will buy, regulators will approve the route on schedule or Year-4 distribution will occur.

Term	Meaning in this report
GMV	Customer transaction value processed. It is not revenue, profit or company cash.
Revenue	Escrow, document/exception and partner/API/ancillary fees after launch discounts.
Gross profit	Revenue less modeled variable and shared COGS.
EBITDA before one-offs	Gross profit less payroll and fixed opex; excludes startup/regulatory one-offs.
Operating result	Gross profit less payroll, fixed opex and modeled one-offs; pre-tax and pre-financing.
Peak cash gap	Largest cumulative monthly operating deficit before financing.
Operating reserve	Spendable liquidity cushion included in the initial ask.
Restricted capital	Regulatory capital held separately; not spendable runway.
Safeguarded client assets	Gross customer escrow balance with an equal liability; 0% available to operations.

Source: Financial Model README, P&L 5yr, Launch Readiness, Cash Flow Statement and Client Funds Memo.

Material limitations

- 1. No operating history.** Deal volume, conversion, churn, CAC, review workload and holding period are assumptions or planning outputs, not observed performance.
- 2. No investment-grade valuation.** The model is a five-year pre-tax, pre-financing operating plan. It does not model corporation tax, interest, depreciation/amortisation, working-capital accounting beyond the cash plan, or an enterprise valuation.
- 3. Year-4 and Year-5 are ambitious.** The base case depends on partner distribution, self-serve onboarding, digital documents and an AI-enabled staffing model.
- 4. Regulatory figures are placeholders.** The VARA application, supervision and restricted-capital assumptions require confirmation from counsel, the partner and regulators.
- 5. Client-fund timing is provisional.** Average safeguarded balances use a 14-day funded-to-release period. A 30-, 60- or 90-day cycle materially increases safeguarding scale but still creates no operating cash.
- 6. Pricing is not willingness-to-pay evidence.** The price book is informed by deal-value and competitor research; only a paid pilot can validate conversion and retention.

Financial conclusion

Blockmediary has a coherent economic design: prefunding removes balance-sheet credit exposure; tiered pricing makes manual complexity pay for itself; digital adoption lowers the take rate without destroying gross margin; and client funds remain segregated from company cash. The model therefore supports a credible claim of potential real value.

The model does not yet support certainty. The base case requires £3.063m, first revenue only after Month 12 and a major Year-4 scale step. The right investor posture is conditional commitment: fund the regulated pilot and production controls, measure the unit economics, and release scale capital only when paid transactions validate price, workload, cycle time, repeat use and distribution.

Bottom line The prototype shows that the escrow can work. The financial model shows what must become true for the business to work.

References and model source map

Ref.	Source
[1]	University of Exeter, BEEM063 Hackathon Main Presentation Brief 2026, p.1.
[2]	Team Transakt, Blockmediary Financial Model - Submission Ready, 13 August 2026.
[3]	Blockmediary, Defensible Average Deal Values for Documentary Escrow, 11 August 2026, especially pp.2-6 and 24-25.
[4]	Blockmediary, DIFC Partner-Led Pilot and VARA Scale Readiness Report, 12 August 2026, especially pp.1-3 and 11.
[5]	Blockmediary, Competitor Analysis, August 2026, especially pp.2-3 and 12-13.
[6]	Blockmediary, Legal and Compliance Risk Assessment, August 2026, especially pp.2-4 and 22-24.
[7]	Blockmediary, Trade Escrow Agreement, August 2026, especially pp.2-4 and 18-19.
[8]	Blockmediary repository README and Business Requirements Document, baselined 9 August 2026.

Primary workbook locations

Topic	Workbook location
Executive headlines and use of funds	Dashboard
Launch timing, regulatory costs and initial funding	Launch Readiness!A5:J23
Tier price book, revenue, cost and contribution	Tier Economics!B3:U35
Five-year operating case and customer bridge	P&L 5yr!C7:G91
Monthly cash flow and break-even	Cash Flow Statement!B9:BI58
Low / Base / High cases	Scenario Engine!A1:R64
Safeguarded client assets	Client Funds Memo!A5:H31
Mechanical validation	Model Checks!A1:G48
Model inputs and source notes	Assumptions; Driver Tables; Risk Register

All financial figures in this report were extracted from the submission-ready workbook dated 13 August 2026 and rounded for presentation. The workbook remains the numerical source of truth.